

## SIMATS ENGINEERING

### ASSESSMENT TOOL 2

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COURSE NAME: COMPUTER NETWORKS  
COURSE CODE: CSA0711

#### COURSE OUTCOME COVERED

##### CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2, BL6)

Assessment Tool Weightage: 30%

#### RUBRICS FOR CALCULATION

| Criteria                     | Weightage | Exemplary (4)                                                                | Proficient (3)                            | Developing (2)                                | Beginning (1)                                  |
|------------------------------|-----------|------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------------------|------------------------------------------------|
| Problem Analysis             | 15%       | Clearly identifies and deeply analyzes the industry problem with all factors | Good understanding with minor gaps        | Basic understanding, some key aspects missing | Poor or incorrect understanding of the problem |
| Solution Design              | 20%       | Innovative, practical, and industry-relevant solution                        | Effective solution with minor limitations | Partially suitable solution                   | Inappropriate or unclear solution              |
| Technical Implementation     | 20%       | Fully functional, accurate, and well-executed implementation                 | Mostly functional with minor errors       | Partially implemented solution                | Incomplete or non-functional implementation    |
| Use of Tools & Technologies  | 10%       | Appropriate and advanced use of relevant tools and technologies              | Correct use of tools                      | Limited or basic use of tools                 | Incorrect or no use of tools                   |
| Problem-Solving Approach     | 10%       | Logical, systematic, and well-structured approach                            | Mostly logical approach                   | Somewhat disorganized approach                | No clear approach                              |
| Innovation & Creativity      | 10%       | Highly creative and original solution                                        | Some creativity                           | Limited originality                           | No creativity                                  |
| Testing & Validation         | 5%        | Thorough testing with real-world scenarios                                   | Adequate testing                          | Minimal testing                               | No testing                                     |
| Documentation & Presentation | 5%        | Clear, professional, and well-documented                                     | Good documentation                        | Basic documentation                           | Poor or no documentation                       |
| Teamwork / Collaboration     | 5%        | Excellent coordination and contribution (if team-based)                      | Good collaboration                        | Limited participation                         | No collaboration or contribution               |

## DETAILED SOLUTIONS — INDUSTRY PROBLEM-SOLVING TASK

Each of the five industry scenarios below is expanded into a complete network-design solution: a problem analysis, the proposed topology with a labelled diagram, a device/media specification, an addressing or configuration scheme where applicable, the technical justification, and a testing and validation plan — directly addressing every rubric criterion (Problem Analysis, Solution Design, Technical Implementation, Use of Tools & Technologies, Problem-Solving Approach, Innovation, Testing & Validation, and Documentation).

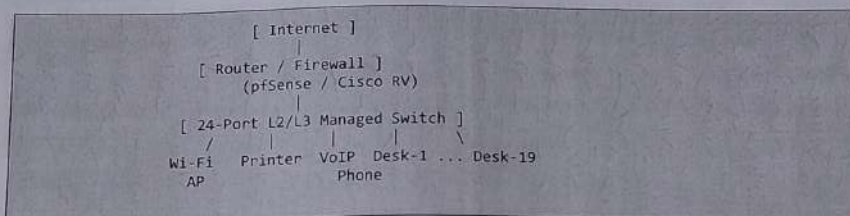
### Solution 1 — Star Topology for a 20-Person Startup

#### 1.1 Problem Analysis

Requirements: 20 employees, a mix of workstations, network printers, and VoIP phones, a tight IT budget typical of a startup, need for simple day-to-day management by a small (often part-time) IT staff, and room to grow to roughly 30 devices without a redesign. Downtime tolerance is moderate — brief single-user outages are acceptable, but a network-wide outage would halt the whole office.

#### 1.2 Proposed Topology and Network Diagram

A pure star topology is proposed, with a single Layer 2/Layer 3 managed switch as the central point of connection for every device, and the switch uplinked to a router/firewall for internet access.



#### 1.3 Device and Cable Specification

| Item              | Model / Type                                                  | Qty      | Purpose                                      |
|-------------------|---------------------------------------------------------------|----------|----------------------------------------------|
| Core switch       | Cisco Catalyst 1300 / Netgear ProSafe, 24-port, L2/L3 managed | 1        | Central connection point; VLAN & QoS capable |
| Router / firewall | Cisco RV series or pfSense appliance                          | 1        | Internet uplink, DHCP, NAT, firewall rules   |
| Wireless AP       | 802.11ac/ax access point                                      | 1-2      | Wi-Fi coverage for laptops & mobiles         |
| Workstations      | Desktop / laptop PCs                                          | 17-19    | End-user devices                             |
| Network printer   | Networked laser/MFP printer                                   | 1-2      | Shared printing                              |
| VoIP phones       | SIP desk phones                                               | up to 20 | Voice traffic (QoS-tagged)                   |
| Patch cabling     | Cat6 UTP, RJ-45                                               | ~22      | Gigabit link, ≤100 m switch-to-device        |

#### 1.4 Sample Cable-Run Table (Switch to Each Device)



| Host      | Central Switch | Cat6 Cable Length (m) |
|-----------|----------------|-----------------------|
| Desk-1    | Switch-Core-A1 | 8.5                   |
| Desk-2    | Switch-Core-A1 | 11.2                  |
| Desk-3    | Switch-Core-A1 | 15.0                  |
| ...       | ...            | ...                   |
| Printer-1 | Switch-Core-A1 | 6.0                   |
| VoIP-1    | Switch-Core-A1 | 9.4                   |
| AP-1      | Switch-Core-A1 | 22.7                  |

All runs <= 100 m -> compliant with IEEE 802.3 / TIA-568

### 1.5 IP Addressing Plan

| VLAN / Pool        | Subnet          | Usable Hosts | Assigned To                        |
|--------------------|-----------------|--------------|------------------------------------|
| Data VLAN 10       | 192.168.10.0/24 | 254          | Workstations & printers            |
| Voice VLAN 20      | 192.168.20.0/24 | 254          | VoIP phones (tagged, QoS priority) |
| Management VLAN 99 | 192.168.99.0/28 | 14           | Switch & AP management interfaces  |

### 1.6 Justification (Fault Isolation & Manageability)

- Fault isolation: because every device has its own dedicated home-run cable to the switch, a single failed cable or NIC affects only that one device — the remaining 19 users are unaffected, unlike a shared-medium design.
- Centralised troubleshooting: link lights and port statistics on the managed switch let a single technician quickly identify which port/cable is faulty, without disturbing other users.
- Growth headroom: a 24-port switch leaves 4-6 spare ports beyond the initial 20 devices, and a second switch can later be daisy-chained (uplinked) if the company grows further, extending the star into an extended star without redesigning the wiring already in place.
- Quality of Service: tagging the VoIP phones into a separate voice VLAN with QoS priority protects call quality even when data traffic (large file transfers) spikes.

### 1.7 Testing and Validation Plan

- Use a cable tester/certifier on every Cat6 run to confirm wire-map, length (<100 m), and Gigabit-capable performance before go-live.
- Verify DHCP scope assignment and connectivity with ping/traceroute from a sample device on each VLAN to the gateway and to the internet.
- Simulate a single cable failure (unplug one workstation) and confirm on the switch's port/LED status and via ping that only that device is affected, validating fault isolation.
- Load-test the VoIP VLAN during a simulated data transfer to confirm QoS priority keeps call quality (jitter/latency) within acceptable limits.

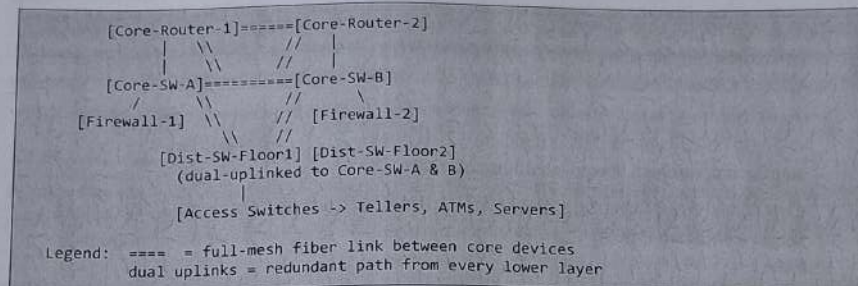
## Solution 2 — Partial Mesh Topology for a Bank

### 2.1 Problem Analysis

Requirements: near-zero downtime for financial transaction processing, regulatory compliance for uptime and data integrity, protection against both link and device failure, low-latency connectivity between data-center devices, and strong resistance to electromagnetic interference given dense equipment racks.

### 2.2 Proposed Topology and Network Diagram

A partial mesh is proposed at the core layer: core switches, core routers, and firewalls are interconnected directly with one another, while distribution and access layers connect upward to at least two core devices for redundancy (dual-homing).



### 2.3 Device and Media Specification

| Item                      | Model / Type                           | Qty        | Purpose                                  |
|---------------------------|----------------------------------------|------------|------------------------------------------|
| Core routers              | Cisco ASR / Juniper MX series (dual)   | 2          | BGP-based redundant internet/WAN edge    |
| Core switches             | Enterprise L3 chassis switches         | 2          | Core mesh; inter-VLAN & OSPF routing     |
| Firewalls                 | Next-gen firewall appliances (HA pair) | 2          | Security, redundant via failover cluster |
| Distribution switches     | L3 switches, dual-uplinked             | per floor  | Aggregation from access layer            |
| Fiber - inter-building    | Single-mode fiber (SMF)                | as needed  | Long-distance, low-loss core links       |
| Fiber - intra-data-center | Multi-mode OM4                         | as needed  | High-bandwidth short links, EMI-immune   |
| Power/backup              | Redundant PSU, UPS, RAID storage       | per device | Continuity during power/disk failure     |

### 2.4 Redundancy and Routing Protocols

| Protocol    | Layer                     | Purpose                                           |
|-------------|---------------------------|---------------------------------------------------|
| OSPF        | Core/Distribution routing | Sub-second rerouting on internal link failure     |
| BGP         | WAN edge (dual ISPs)      | Redundant, policy-based internet connectivity     |
| HSRP / VRRP | Core gateway redundancy   | Virtual default gateway shared by Core-SW-A/B     |
| STP / RSTP  | Loop prevention           | Prevents broadcast storms on redundant mesh links |

### 2.5 Justification



- No single point of failure: every core device has at least two independent paths to every other core device, so one failed link or node cannot isolate any part of the network.
- Millisecond failover: OSPF/BGP continuously recompute shortest paths; when a monitored link goes down, traffic reroutes automatically without manual intervention, meeting the bank's near-zero-downtime requirement.
- Fiber over copper: single-mode/multi-mode fiber eliminates electromagnetic interference from banking hall equipment (UPS units, HVAC) and supports the very high bandwidth financial transaction systems require.
- Regulatory alignment: redundant power, RAID storage, and dual-path networking collectively support the uptime and data-integrity guarantees that banking regulators (e.g., RBI/PCI-DSS style frameworks) commonly require.

#### *2.6 Testing and Validation Plan*

- Perform planned link-failure drills (physically disconnecting one core fiber link) and measure failover time using continuous ping/traceroute — target sub-second to low-second convergence.
- Validate BGP failover by simulating an ISP outage and confirming automatic route withdrawal/re-advertisement to the secondary ISP.
- Run a full disaster-recovery test including firewall HA failover and RAID rebuild simulation to confirm data integrity is preserved.
- Conduct periodic penetration testing and latency benchmarking between branch/distribution switches and the core to ensure transaction-processing SLAs are met.

### Solution 3 — Bus Topology for a Budget Classroom Lab

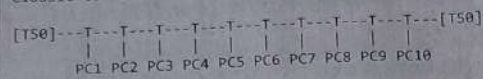
#### 3.1 Problem Analysis

Requirements: minimum possible hardware cost, a small number of computers (typically 10-20) in one room, supervised light usage (web browsing, file sharing, no mission-critical services), and no need for a dedicated server.

#### 3.2 Proposed Topology and Network Diagram

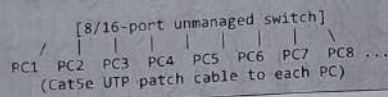
Two implementation options are given: the classic coaxial bus, and a modern low-cost variant using a single unmanaged switch (recommended for easier sourcing of parts today).

Classic coaxial bus:



T50 = 50-ohm terminator T = T-connector (BNC)

Modern low-cost variant:



#### 3.3 Bill of Materials (approx., 16 seats)

| Item                             | Type                                     | Qty   | Approx. Notes                   |
|----------------------------------|------------------------------------------|-------|---------------------------------|
| Backbone cable                   | RG-58 coax (or single Cat5e run)         | 1 run | Length of room                  |
| T-connectors + BNC               | Coaxial T-piece                          | 16    | One per PC (classic option)     |
| Terminators                      | 50-ohm end caps                          | 2     | One at each end of the bus      |
| Unmanaged switch (modern option) | 8- or 16-port 10/100/1000                | 1     | Replaces terminators + hub role |
| Patch cables                     | Cat5e UTP                                | 16    | PC to switch (modern option)    |
| Teacher PC                       | Shares files/printer via OS file-sharing | 1     | Acts as informal server         |

#### 3.4 Justification

- Lowest cost: eliminating a managed switch or server removes the two most expensive components of a small network, keeping the design within a tight classroom budget.
- Adequate for the use case: light, supervised, non-critical traffic (browsing, shared files) tolerates the occasional collision and does not need enterprise-grade reliability.
- Simple to deploy and understand: the flat, single-segment design is also pedagogically useful, letting students literally see how a shared-medium bus network works.

#### 3.5 Limitations and Mitigations

- Single point of failure (whole bus down on one break) — mitigated by keeping spare coax/patch cable and terminators on hand for quick replacement, and by preferring the

unmanaged-switch variant, which isolates a single bad cable to one PC instead of the whole segment.

- Collision-prone under heavy load — acceptable given the light, supervised usage pattern described; if usage grows, the room can be upgraded to a star topology later using the same PCs.

### 3.6 Testing and Validation Plan

- Test backbone continuity and the 50-ohm terminator resistance with a multimeter/cable tester before connecting any PCs.
- Verify each PC can reach the teacher's shared folder and the internet gateway using ping and a simple file-copy test.
- Deliberately disconnect one T-connector/patch cable to confirm the expected failure behaviour (whole bus down for coax, single PC down for the switch variant), documenting the result for student learning.



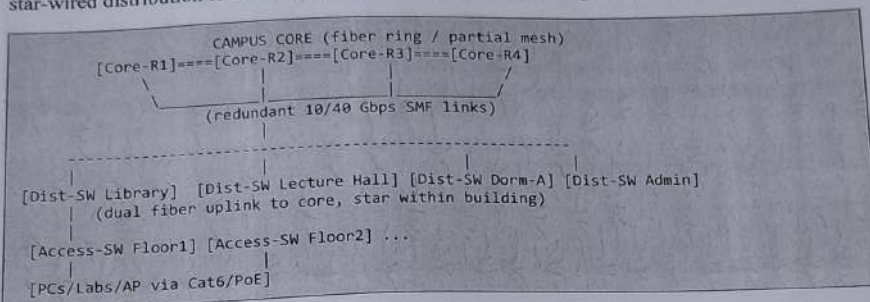
## Solution 4 — Hybrid Topology for a University Campus

### 4.1 Problem Analysis

Requirements: thousands of users across many buildings (lecture halls, labs, libraries, dorms, admin), high redundancy needs at the backbone, manageable wiring within each building, cost-efficient labs, wireless coverage campus-wide, and painless future expansion as new buildings are added.

### 4.2 Proposed Topology and Network Diagram

A three-layer hierarchical hybrid design is proposed: a fiber ring/partial-mesh backbone (core), star-wired distribution to each building, and star/tree access wiring within labs and offices.



### 4.3 Layered Device Specification

| Layer        | Devices                                                   | Media                         | Role                                                        |
|--------------|-----------------------------------------------------------|-------------------------------|-------------------------------------------------------------|
| Core         | Core routers / multilayer switches (ring or partial mesh) | Single-mode fiber, 10/40 Gbps | Campus backbone; no single link isolates a building         |
| Distribution | One L3 distribution switch per building, dual-uplinked    | Fiber to core                 | Aggregates each building; boundary for building-level VLANs |
| Access       | Floor-level access switches                               | Cat6 UTP (and PoE for APs)    | Star wiring to end devices within a building                |
| Wireless     | PoE-powered access points                                 | PoE over Cat6                 | Campus-wide Wi-Fi without separate power cabling            |

### 4.4 Justification

- Redundant backbone: the ring/partial-mesh core means a single fiber cut between two buildings reroutes automatically without isolating either building, satisfying the scale and uptime needs of a large campus.
- Manageable buildings: star wiring inside each building keeps troubleshooting simple for the local IT team, exactly as in Solution 1, but scoped to one building instead of the whole campus.



- Cost-efficient labs: bus/tree wiring inside individual labs (or simple star from a cheap access switch) keeps lab costs low while still uplinking into the robust building/core structure.
- Painless expansion: adding a new building only requires a new distribution switch and a fiber run to the core — the industry-standard reason the three-layer (core/distribution/access) model is used at almost every large enterprise and campus.
- PoE for wireless: powering access points over the same Cat6 cable that carries data removes the need for electricians to run separate power lines to ceiling-mounted APs.

#### 4.5 Testing and Validation Plan

- Test backbone redundancy by disabling one core fiber link and confirming automatic rerouting (via routing-protocol convergence time measurement) with no building losing connectivity.
- Verify each building's dual uplinks by disabling one uplink and confirming the distribution switch fails over to the second without manual reconfiguration.
- Load-test Wi-Fi coverage and PoE budget in a sample building during peak class-change hours to confirm access points remain powered and responsive.
- Run end-to-end connectivity/performance tests (ping, traceroute, iperf-style throughput tests) from a sample of access-layer devices in different buildings to the core and to the internet edge.

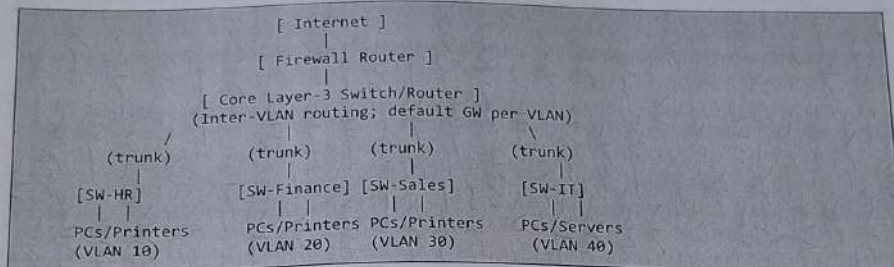
## Solution 5 — Multi-Department Network with VLANs and Inter-VLAN Routing

### 5.1 Problem Analysis

Requirements: several departments (HR, Finance, Sales, IT) that each need internal communication, controlled communication between departments, reduced broadcast traffic, strong security/segregation (e.g., Finance data should not be broadcast to Sales), internet access with firewall protection, and a design that scales as more departments or staff are added.

### 5.2 Proposed Topology and Network Diagram

Each department is wired in a star to its own access-layer Layer 2 switch. All access switches trunk up to a central Layer 3 switch/router that performs inter-VLAN routing, with a firewall router connecting the whole network to the internet.



### 5.3 VLAN and IP Addressing Plan

| VLAN ID | Department | Subnet          | Default Gateway |
|---------|------------|-----------------|-----------------|
| VLAN 10 | HR         | 192.168.10.0/24 | 192.168.10.1    |
| VLAN 20 | Finance    | 192.168.20.0/24 | 192.168.20.1    |
| VLAN 30 | Sales      | 192.168.30.0/24 | 192.168.30.1    |
| VLAN 40 | IT         | 192.168.40.0/24 | 192.168.40.1    |
| VLAN 99 | Management | 192.168.99.0/28 | 192.168.99.1    |

### 5.4 Representative Switch/Router Configuration (Cisco IOS style)

```
! --- Access switch (e.g., SW-HR) ---
vlan 10
 name HR
interface range fastEthernet 0/1-20
 switchport mode access
 switchport access vlan 10
interface gigabitEthernet 0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,99

! --- Core Layer-3 switch/router ---
vlan 10
 name HR
vlan 20
 name FINANCE
```



```

vlan 30
 name SALES
vlan 40
 name IT
!
interface vlan 10
 ip address 192.168.10.1 255.255.255.0
interface vlan 20
 ip address 192.168.20.1 255.255.255.0
interface vlan 30
 ip address 192.168.30.1 255.255.255.0
interface vlan 40
 ip address 192.168.40.1 255.255.255.0
!
ip routing
! enables inter-VLAN routing

```

### 5.5 Justification

- Security through segmentation: placing each department in its own VLAN means a broadcast or a compromised device in Sales cannot directly reach Finance or HR traffic; inter-VLAN access can be explicitly restricted with access-control lists on the core router.
- Reduced broadcast traffic: broadcasts (ARP, DHCP discover, etc.) stay within a department's own VLAN/broadcast domain instead of flooding the entire company, improving overall performance as headcount grows.
- Centralised routing and internet access: the core Layer-3 switch/router handles all inter-department traffic and hands off to the firewall router for internet access, giving one clear choke point for applying security policy.
- Scalability: adding a new department is simply a matter of creating a new VLAN, assigning a subnet, and trunking a new (or existing) access switch — no rewiring of the existing departments is required.
- Redundancy-ready: redundant uplinks/backup paths (e.g., a second core switch with HSRP/VRRP) can be layered onto this same design without changing the department-facing access layer.

### 5.6 Testing and Validation Plan

- Confirm VLAN isolation by attempting to ping between two hosts in different VLANs before enabling inter-VLAN routing — the ping should fail, proving broadcast-domain separation.
- Enable inter-VLAN routing and re-test connectivity between departments that are permitted to communicate (e.g., IT to all departments for support), confirming routing works as intended.
- Verify ACLs by attempting a denied cross-department connection (e.g., Sales to Finance) and confirming it is blocked while permitted traffic (e.g., HR to IT helpdesk) passes.